Enhanced Mask2Former With Multi-Scale and High-Resolution for Concrete Bridge Crack Semantic Segmentation

Di Wang, Lu Dai, Fengquan Song, Han Yang, Jinhong Li, Xianyi Yang, and Shixin Jiang

***Abstract*— Bridge cracks can threaten transportation safety and infrastructure, leading to severe failures if not detected. Timely, accurate detection is vital for bridge safety and preventing costly repairs. Traditional methods rely on manual inspections and basic image processing, making them slow, error-prone, and unable to detect fine cracks or handle noise. Current semantic segmentation methods face challenges in accurately identifying cracks, particularly in maintaining segmentation continuity and handling localized crack features. This paper proposes an improved bridge crack segmentation algorithm based on Mask2Former, focusing on intelligent automation in bridge maintenance by preserving segmentation continuity and handling diverse crack morphologies. The Mask2Former architecture is enhanced through the introduction of a multi-scale feature fusion module, which utilizes a fusion block to process concatenated feature maps via RepBlock, strengthening feature representation, particularly for small cracks. Furthermore, a high-resolution feature processing module is integrated, employing wavelet transform convolution (WTConv) with multi-level wavelet transformations for extensive receptive field coverage. WTConv is refined with detail enhancement convolution (DEConv), which maintains parameter efficiency, along with a channel and spatial attention mechanism to better capture fine details, thus improving the segmentation of localized cracks and effectively handling background noise. Experiments on benchmark datasets demonstrate that the proposed algorithm achieves mIoU of 86.33%, F1-Score of 85.07%, and pixel accuracy of 98.70%, surpassing existing methods in fine crack detection. Ablation experiments show that both MSFM and HRFPM significantly outperform the baseline, achieving mIoU scores of 86.22% and 85.99%, and F1-scores of 84.95% and 84.65%, respectively. These results highlight the effectiveness of the proposed method in improving the robustness, accuracy, and efficiency of bridge crack segmentation, with significant potential for enhancing bridge health monitoring in transportation infrastructure.**

Received 7 January 2025; revised 29 May 2025; accepted 5 June 2025. Date of publication 23 June 2025; date of current version 21 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62103068, Grant 52478141, and Grant 62003063; in part by the Science and Technology Research Program of Chongqing Municipal Education Commission of China under Grant KJQN202100745 and Grant KJZD-K202400709; in part by the Natural Science Foundation of Chongqing, China, under Grant CSTB2022NSCQ-MSX1599; and in part by the Graduate Student Research Innovation Project of Chongqing Jiaotong University under Grant 2024s0103. The Associate Editor for this article was C. Ahlstrom. *(Corresponding author: Di Wang.)*

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Digital Object Identifier 10.1109/TITS.2025.3578810

***Index Terms*— Bridge crack detection, semantic segmentation, convolutional neural network, Mask2Former, feature fusion.**

I. Introduction

As a key component of economic development and daily transportation, bridges play an irreplaceable role in modern infrastructure. By the end of 2023, the total number of railway and highway bridges in China exceeded one million. Among them, highway bridges accounted for 1.0793 million, an increase of 46,100 from the previous year. This included 10,239 extra-large bridges with a total length of 18.7301 million linear meters, and 177,700 large bridges [1]. Over time, concrete bridges are prone to damage such as crack propagation and cavity formation, driven by environmental erosion, load effects, and material aging. These issues can accelerate material degradation and reduce structural capacity, potentially leading to catastrophic failure. For example, the collapse of Seongsu Bridge in Seoul in 1994, caused by fatigue cracks and poor maintenance, resulted in 32 deaths and 17 injuries [2]. Similarly, in 2021, a bridge on the Huyu Expressway in Ezhou, China, failed due to cracking, causing 4 deaths, 8 injuries, and economic losses of 30.18 30.18 million RMB [3]. The structural health of bridges directly affects public safety and economic stability. Cracks are key indicators of early-stage damage; if not addressed promptly, they can lead to severe consequences [4], [5]. Traditionally, bridge crack detection relies mainly on manual inspection, which is not only time-consuming and labor-intensive but also susceptible to subjective biases, leading to risks of misjudgment and missed detection [6]. In recent years, computer vision technologies have gained prominence in crack segmentation due to their cost-effectiveness, rapid detection capabilities, and broad applicability [7], [8]. Early research predominantly focused on traditional image processing techniques, including threshold segmentation, edge detection, and region-growing algorithms [9]. Threshold segmentation divides images by grayscale thresholds, offering computational efficiency but suffering from noise sensitivity. Lu et al. [10] proposed a local adaptive threshold strategy to enhance crack recognition, yet its performance remains heavily dependent on manual threshold selection and degrades under complex noise. Edge detection techniques, such as the improved canny operator by Zhao et al. [11] leverage gradient features to extract

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structural contours. While robust to lighting variations, these methods exhibit high sensitivity to noise and require empirical parameter tuning, often resulting in discontinuous edges or artifacts. Compared to edge detection methods [12], region-growing segmentation divides images into regions via pixel similarity criteria, utilizing an iterative seed-based growth mechanism that excels in processing blurred edges and irregular geometries. However, as demonstrated by Li et al. [13] such algorithms remain susceptible to over-segmentation due to their inherent reliance on initial seed selection and constrained parameter adaptability. Although traditional image processing methods exhibit high detection accuracy in specific scenarios, their generalization ability is limited by inherent flaws such as insufficient algorithm robustness, limited ability to characterize small-scale cracks, and experience-driven parameter configuration. These limitations lead to a significant decline in detection performance in engineering practice, highlighting the urgent need to develop an adaptive algorithmic framework for breakthrough improvements.

With the rapid development of deep learning technology, methods such as object detection [14] and semantic segmentation [15] have shown potential in image processing tasks. Compared to traditional image processing algorithms, applying deep learning techniques to crack segmentation tasks offers significant advantages [16]. Deep learning methods not only automate feature extraction, eliminating the need for cumbersome manual intervention and improving detection efficiency, but also achieve high accuracy and robustness, allowing models to capture crack details in complex environments accurately and adapt to variations in lighting, materials, and backgrounds. Consequently, deep learning has garnered widespread attention and has become a crucial direction for breakthroughs in bridge crack detection. Object detection frameworks like YOLO-MF [17] and YOLOv4-FPM [18] achieved accuracies around 98% through architectural optimizations, however, anchor-based mechanisms have limitations in detecting slender cracks. The feature extraction of such cracks is easily affected by background interference, resulting in localization errors Lau et al. [19] demonstrated efficacy of U-Net in automating feature extraction, but its limited receptive field compromises global context modeling, leading to an F1-Score lower than transformer-based methods [20] in cluttered backgrounds. The methods above, based on convolutional neural networks (CNNs), have improved crack detection through structural optimization and algorithm integration. However, they are limited by local receptive fields of CNN, resulting in poor representation of long-range dependencies and blurred boundaries in complex backgrounds. To address these limitations, subsequent research has explored integrating transformer architectures for global perceptual capabilities. Fang et al. [21] proposed an enhanced crack detection framework that combined TransUNet with external attention to improve robustness and structural representation. Qi et al. [22] proposed VCV-Net, a hybrid framework that integrated vision transformer (ViT) and level set theory for precise bridge pavement defect segmentation. It employed two parallel decoders under a unified objective function to address challenges like variable defect shapes and blurred boundaries. Recent hybrid architectures, such as Mix-Graph CrackNet [23] and DTrC-Net [24], combine CNNs and Transformers to balance local and global features. Despite the progress made in improving the efficiency and accuracy of crack segmentation using the above methods, there are still challenges in practical engineering applications. Crack images typically have continuous and narrow morphological features, and the distribution of cracks and background pixels is often extremely uneven, which makes it difficult for the model to accurately segment the cracks.

Fig. 1. Dataset images.

To address these issues, this study proposes a bridge crack semantic segmentation method based on the improved Mask2Former to further improve the segmentation effect in real-world applications. Mask2Former [25] was a general image segmentation architecture that could handle any image segmentation task, including semantic segmentation. The core component of Mask2Former was its masked attention mechanism, which limited cross-attention only to the predicted masked regions to extract local features. This not only reduced the computational complexity but also improved the training efficiency and performance of the model.

II. Dataset

To evaluate the performance of this method across diverse scenarios and ensure the generalization capability and practical applicability of the model, a comprehensive dataset was constructed, consisting of 200 self-collected images obtained through on-site photography and manual annotation, along with 1,300 images from the publicly available SDNET2018 [26] dataset—a widely recognized benchmark for concrete surface defect analysis. This combined dataset (total 1,500 images) was designed to capture various bridge surface defects, including fine cracks obscured by interfering elements such as weeds, shadows, and graffiti. For the experiments, these 1,500 representative images of concrete bridge surface cracks (illustrated in Fig. 1) were utilized and divided into a training set (70%) and a testing set (30%). To further enhance robustness, the training set was expanded using data augmentation techniques. The final dataset thus covers a wide range of concrete crack scenarios, including cracks of varying widths and challenging backgrounds featuring shadows, surface roughness, scaling, edges, holes, and debris.

The data was labeled at the pixel level using the LabelMe annotation tool, where each pixel in the original image was divided into two categories. Red pixels were classified as

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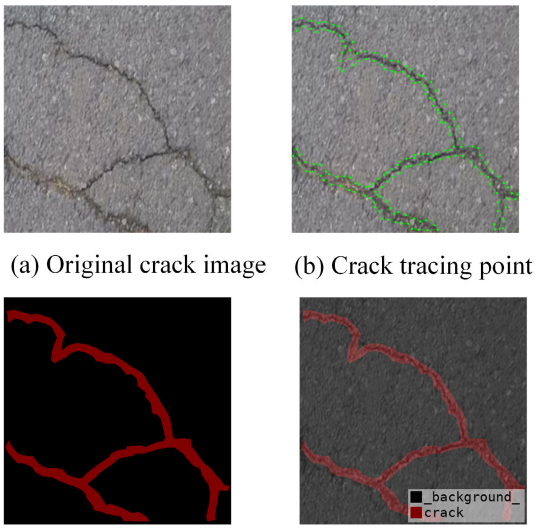

(a) Original crack image (b) Crack tracing point

(c) Crack label (d) Mask and instance

Fig. 2. Production of data labels.

crack-related, while black pixels were classified as background. The process of creating the dataset is shown in Fig. 2.

III. Method

In this section, we present a detailed explanation of the improvements made to the Mask2Former model to enhance its performance in the bridge crack detection task, as depicted in Fig. 3. The original Mask2Former is a general segmentation model based on the Transformer architecture. To improve its performance and adapt it to the complex task of bridge crack segmentation, we designed a multi-scale fusion module (MSFM) and introduced a fusion block [27] to concatenate features along the channel dimension. Feature map processing is then performed using 1×1 convolutional layers and RepBlock, facilitating the fusion of deep and shallow convolutional features. This enables the model to retain both high-level global information and low-level detailed information, thereby improving segmentation accuracy. Additionally, we developed a high-resolution feature processing module (HRFPM) that leverages wavelet [28] to perform multi-level decomposition of the input features and capture multi-scale information through convolution. These multi-level convolutional outputs are then merged via inverse wavelet transform, enhancing the feature of model representation capabilities. To further improve the ability of the model to capture fine details, we introduced detail enhancement convolution (DEConv) [29], which is designed to extract finer crack features. At the output stage of the wavelet transform, the channel and spatial attention mechanism (CBAM) [30] is incorporated to focus on essential features in both the channel and spatial dimensions, effectively boosting the feature of model expression capability.

A. Mask2Former

Mask2Former was a novel semantic segmentation model based on the Transformer architecture, integrating the technical strengths of object detection and semantic segmentation to simultaneously address the tasks of instance segmentation, semantic segmentation, and panoptic segmentation. The Mask2Former architecture consisted of a backbone network, a pixel decoder, and a transformer decoder. The backbone network output four feature maps, which served as inputs for further processing by the pixel decoder and transformer decoder. The pixel decoder employed multi-scale deformable attention (MSDeformAttn) to process feature maps at scales of 1/8, 1/16, and 1/32, and generated a 1/4-scale feature map through an upsampling process. The transformer decoder was the core component of the Mask2Former architecture, responsible for producing the final classification and mask predictions. Its mask attention mechanism differed from traditional global cross-attention by confining attention computation to the predicted mask area, thereby enhancing both efficiency and accuracy. This approach mitigated the limitation of traditional global cross-attention, which required extensive training to learn how to focus on local object regions. Mathematically, the implementation of the mask attention mechanism was achieved through the following operations

$$\mathbf{X}_l = \text{softmax}(\mathcal{M}_{l-1} + \mathbf{Q}_l\mathbf{K}_l^{\mathrm{T}})\mathbf{V}_l + \mathbf{X}_{l-1} \tag{1}$$

where $\mathbf{X}_l$ represented the output feature of layer l corresponding to the updated query of the current feature. $\mathbf{Q}_l$ was the query matrix derived from the previous layer $\mathbf{X}_{l-1}$. $\mathbf{K}_l$ and $\mathbf{V}_l$ are the key and value matrices representing the input image features. $\mathcal{M}_{l-1}$ was the binary mask from the previous layer. The softmax was used to normalize the attention scores.

$$\mathcal{M}_{l-1}(x, y) = \begin{cases} 0 & \text{if } \mathbf{M}_{l-1}(x, y) = 1 \\ -\infty & \text{otherwise} \end{cases} \tag{2}$$

Equation (2) represented the mask construction process. For the position (x, y) of each pixel, it was judged based on the mask value. If the value at position $\mathcal{M}_{l-1}$ is equal to 1, indicating that the pixel belongs to the foreground region, the value at this position was maintained as 0. If the value at this position was not 1, meaning it belonged to the background region, it was assigned as $-\infty$. In the subsequent attention calculation, positions with a value of 0 in the mask matrix were retained, while positions with a value of $-\infty$ were completely ignored through the softmax operation. This ensured that the model focuses only on the foreground region.

B. Multi-Scale Fusion Module

To effectively integrate low-level and high-level features, the original Mask2Former model employed a straightforward upsampling strategy, elevating the low-resolution feature map (1/8) to a higher resolution (1/4), and then concatenating it with the 1/4 feature map from the backbone. This operation generates a 1/4 resolution feature map that provides a richer feature representation for each pixel, thereby facilitating the subsequent crack segmentation task. However, significant disparities exist in spatial resolution and feature representation between the low-resolution and high-resolution feature maps. Direct concatenation may result in incompatible information, which could affect the efficacy of the fused features. The low-resolution feature map loses some high-frequency details

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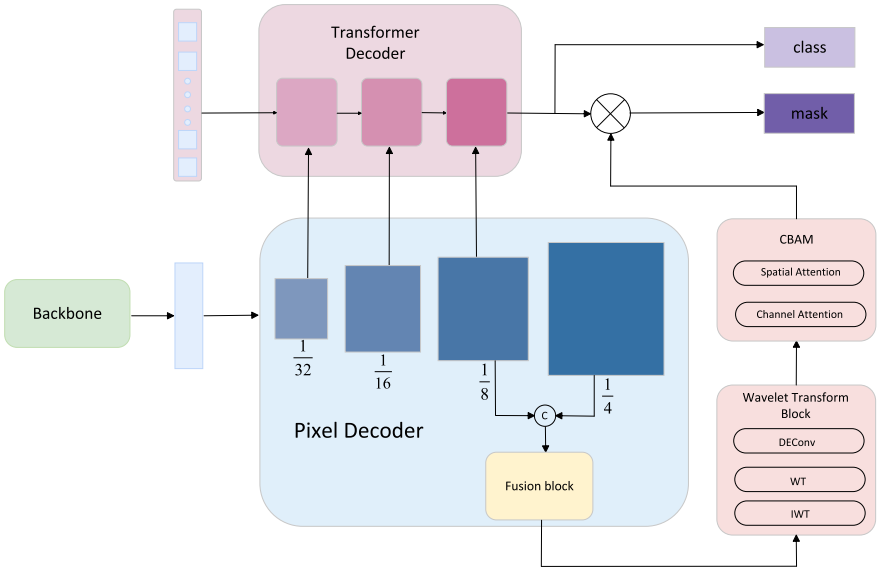

Fig. 3. Overall architecture of the proposed method.

during processing, and direct concatenation is unable to compensate for the absence of these details. This may result in a blurred or incomplete generated feature map, failing to fully leverage the advantages of features at different resolutions, thereby affecting the overall performance of the model. To address this, a feature fusion module was designed, and a fusion block was introduced, as shown in Fig. 4. This module initially concatenates the two feature maps with a channel count of C along the channel dimension to generate a feature map with a channel count of 2C. Next, the concatenated feature map is processed through two 1×1 convolutional layers to adjust its channel count. One set of feature maps is passed through the RepBlock, where feature extraction is performed by combining 3×3 and 1×1 convolutional layers. This further enhances feature representation, particularly in capturing the complex features of bridge cracks. The feature map processed by the RepBlock is then added element-wise to the feature map that did not pass through the RepBlock, thereby fusing the deep convolutional and shallow convolutional features. This fusion allows the model to retain global and detailed information about the cracks. When applying this module to the crack segmentation task, by concatenating and fusing features of different scales in the channel dimension, the model can simultaneously capture the detailed features and overall structure of the cracks. This is extremely crucial for dealing with cracks of diverse sizes and morphologies that occur in crack segmentation and can effectively enhance the segmentation performance of the model.

Compared to traditional multi-scale fusion approaches such as direct concatenation of U-Net or pyramid pooling of PSPNet, the proposed MSFM introduces adaptive feature interaction RepBlock. While methods like FCN and DeepLabv3+ rely on static upsampling or dilated convolutions, which may lead to incompatible feature representations between deep and

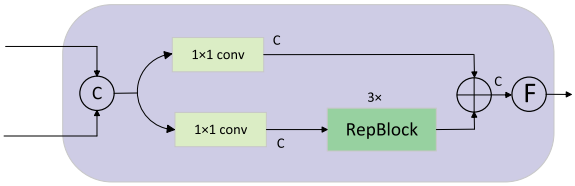

Fig. 4. Multi-scale fusion module.

shallow layers, MSFM dynamically adjusts the fusion process to ensure compatibility between high-level semantics and low-level details. Furthermore, unlike attention-based fusion techniques that require heavy computational overhead, MSFM optimizes parameter efficiency by merging 3×3 and 1×1 convolutions during inference, achieving robust segmentation with reduced complexity.

C. High-Resolution Feature Processing Model

In the original Mask2Former, the final mask prediction is jointly determined by the query vector in the Transformer decoder and the feature map at quarter resolution. The high-resolution feature map preserves fine details in the image, such as crack boundaries, textures, and shapes, ensuring the visibility of small objects and enhancing the ability of model to recognise and segment these objects. However, further processing of the high-resolution feature map may improve segmentation accuracy. Therefore, a high-resolution feature processing module was designed, as illustrated in Fig. 5. In this module, WTConv was introduced. First, the input image is decomposed into subbands of different frequencies via wavelet transform (WT) and small-sized convolution kernels are applied to these subbands to capture multi-scale information. The convolved subbands are then merged using the

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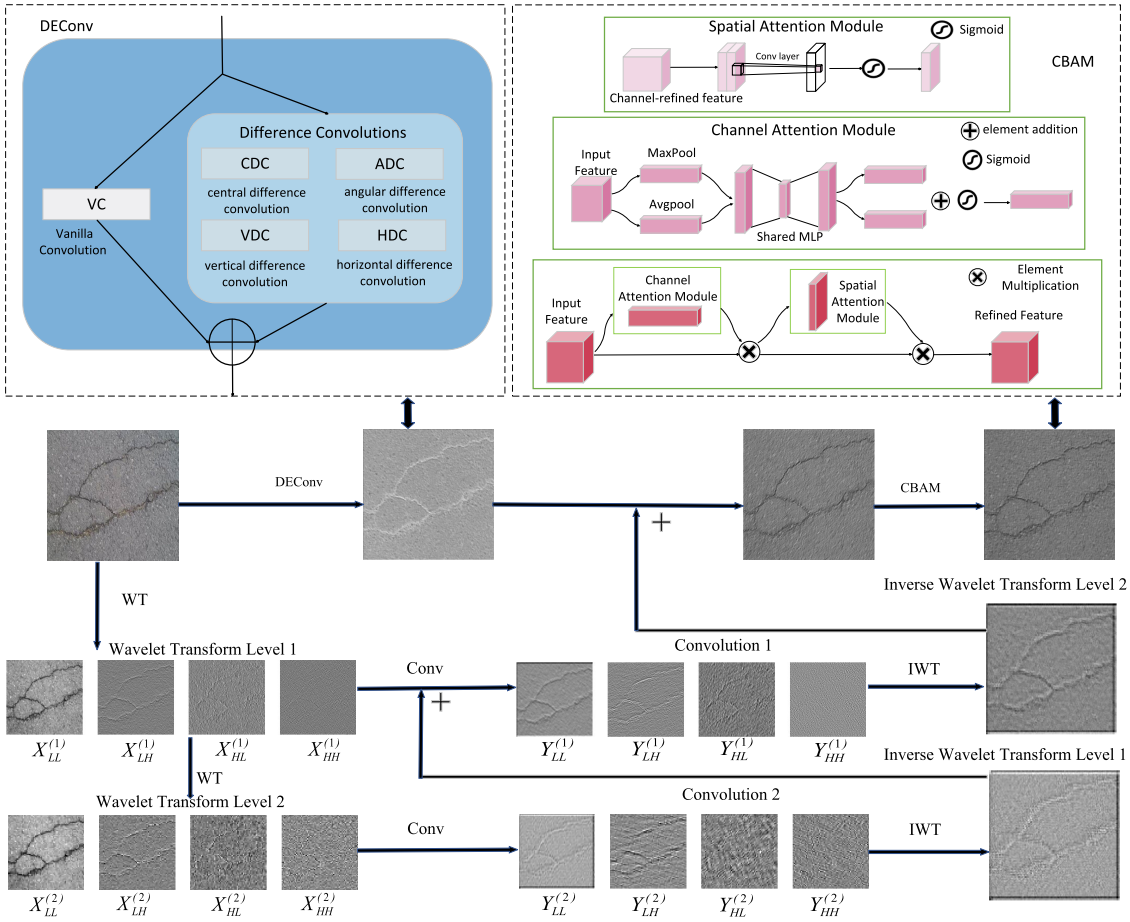

Fig. 5. High-resolution feature processing model.

inverse wavelet transform (IWT) to generate the final output, achieving a large receptive field coverage of the original input. This process can be mathematically expressed as follows,

$$X_{LL}^{(1)}, X_{LH}^{(1)}, X_{HL}^{(1)}, X_{HH}^{(1)} = \mathrm{WT}(X) \tag{3}$$

$$Y_{LL}^{(1)}, Y_{LH}^{(1)}, Y_{HL}^{(1)}, Y_{HH}^{(1)} = \mathrm{Conv}(W^{(1)}, (X_{LL}^{(1)}, X_{LH}^{(1)}, X_{HL}^{(1)}, X_{HH}^{(1)})) \tag{4}$$

$$X_{LL}^{(2)}, X_{LH}^{(2)}, X_{HL}^{(2)}, X_{HH}^{(2)} = \mathrm{WT}(X_{LL}^{(1)}) \tag{5}$$

$$Y_{LL}^{(2)}, Y_{LH}^{(2)}, Y_{HL}^{(2)}, Y_{HH}^{(2)} = \mathrm{Conv}(W^{(2)}, (X_{LL}^{(2)}, X_{LH}^{(2)}, X_{HL}^{(2)}, X_{HH}^{(2)})) \tag{6}$$

$$Z^{(2)} = \mathrm{IWT}(Y_{LL}^{(2)}, Y_{LH}^{(2)}, Y_{HL}^{(2)}, Y_{HH}^{(2)}) \tag{7}$$

$$Z^{(1)} = \mathrm{IWT}(Y_{LL}^{(1)} + Z^{(2)}, Y_{LH}^{(1)}, Y_{HL}^{(1)}, Y_{HH}^{(1)}) \tag{8}$$

Equation (3) illustrates the implementation of a first-level wavelet transformation on the input X, resulting in four components $X_{LL}^{(1)}$,$X_{LH}^{(1)}$,$X_{HL}^{(1)}$, and $X_{HH}^{(1)}$. In these notations, the first letter in the subscript denotes the horizontal component, and the second letter denotes the vertical component. Specifically, L indicates the low-frequency information, while H indicates the high-frequency information. Thus, $X_{LL}^{(1)}$ represents the low-frequency component, while $X_{LH}^{(1)}$,$X_{HL}^{(1)}$, and $X_{HH}^{(1)}$ correspond to the high-frequency components in the vertical, horizontal, and diagonal directions, respectively. W represents the weights of the convolutional layers, Z represents the reconstructed feature map at different levels after applying the IWT. Based on WTConv, we have refined the ordinary convolution and introduced the DEConv. DEConv encompasses five convolutional layers (four differential convolutions and one common convolution), which handle the input features in a parallel fashion, thereby better capturing the detailed features within the images. Through the parameter reconstruction technique, DEConv is capable of fully exploiting the feature extraction advantages of multiple convolutions without augmenting the inference cost. Furthermore, before generating the final output, the CBAM attention mechanism was integrated into the process. CBAM consists of two key components, the channel attention module and the spatial attention module. By focusing on critical features in both the channel and spatial dimensions simultaneously, it enhances the representational capacity of the extracted features. This mechanism can be mathematically described as follows,

$$Z_{\mathrm{DEconv}} = \mathrm{DEConv}(W_{\mathrm{DEConv}}, Z^{(1)}) \tag{9}$$

$$Z_{\mathrm{CBAM}} = \mathrm{CBAM}(Z_{\mathrm{DEConv}}) \tag{10}$$

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TABLE I
EXPERIMENTAL ENVIRONMENT CONFIGURATION

Type	Statement
Operating System	Windows 10
RAM	32G
CPU	Intel I5-13490F
GPU	NVIDIA GeForce RTX 4060 Ti
CUDA version	12.1
Pytorch version	2.0.1

The HRFPM overcomes limitations in existing high-resolution processing methods. While dilated convolutions or encoder-decoder frameworks struggle to capture thin cracks and suppress background noise, HRFPM uniquely integrates WTConv and DEConv. This combination explicitly decomposes features into multi-scale subbands to retain structural continuity while enhancing fine crack edges. Additionally, unlike transformer-based approaches that solely rely on global attention, HRFPM incorporates CBAM to prioritize critical spatial and channel features post-reconstruction, effectively mitigating interference from shadows and debris. By preserving high-frequency details through inverse wavelet transforms, HRFPM avoids the fragmented edges common in upsampling-based methods, ensuring precise segmentation even in complex scenarios.

IV. EXPERIMENT

A. Experimental Configuration

The experiment is based on the PyTorch deep learning framework, employing the AdamW optimizer with a weight decay of 0.05, a learning rate of 0.0001, and 80,000 iterations. Table I presents a comprehensive overview of the experimental configuration.

B. Evaluation Metrics

To objectively evaluate the performance of the trained model in crack segmentation, In the experimental process, three evaluation metrics were used, namely mIoU, Pixel Accuracy (PA), and F1-Score.

mIoU is a commonly used evaluation metric in semantic segmentation tasks, representing the average of the IoUs across all classes. It is employed to measure the proportion of overlapping regions between the predicted result and the ground truth. PA represents the ratio of correctly predicted pixels to the total number of pixels and is used to assess the overall prediction accuracy of the model for the entire image. The F1-Score is a commonly used evaluation metric for measuring the predictive accuracy of the model at the pixel level. It combines Precision and Recall to evaluate the balance of pixel predictions for each class. The relevant formulas are presented below,

$$\mathrm{IoU} = \frac{TP}{TP + FP + FN} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{IoU}_i \tag{12}$$

$$\mathrm{PA} = \frac{TP + TN}{TP + FP + TN + FN} \tag{13}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{14}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{15}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{16}$$

where TP represents the number of pixels that the model correctly identifies as cracks, FP represents the number of pixels that the model incorrectly identifies as cracks, FN represents the number of actual crack pixels that the model fails to identify as cracks, and TN represents the number of pixels that the model correctly identifies as non-cracked. N represents the total number of categories.

C. Analysis of the Comparative Experimental Results

To validate the effectiveness of the concrete bridge crack identification method based on the improved Mask2Former, it was compared with several mainstream semantic segmentation algorithms. These included Deeplabv3+, which was a semantic segmentation network that integrated dilated convolution and an encoder-decoder structure, PSPNet, which captured global context information through a pyramid pooling module, FCN, known for its rich architecture and large number of parameters, SegFormer, a transformer-based network combining hierarchical transformer encoders with lightweight multilayer perceptron decoders for efficient semantic segmentation, K-Net, a kernel-based segmentation framework employing dynamic kernels for flexible feature fusion and mask prediction, and DPT, a vision transformer architecture specialized in dense prediction tasks utilizing transformer modules adapted for depth estimation and semantic segmentation. In the process of training the mode, we recorded the changes in mIoU and F1-Score over time for various networks. As shown in Fig. 6, the improved approach demonstrates clear superiority over other methods in terms of mIoU and F1-Score. In terms of mIoU, the improved approach shows a rapid increase in the early iteration phases, and in the middle and later stages, its performance remains stable, with a smooth and steadily ascending curve. In contrast, although Mask2Former performs comparably to the improved approach during the initial iterations, its performance fluctuates as training progresses, particularly during the middle iterations, where the rate of mIoU improvement slows down, and its final performance is slightly inferior to the improved approach. Other methods, such as FCN and PSPNet, exhibit considerable fluctuations throughout the training process, especially during the middle iterations, where the mIoU value declines significantly, indicating weaker stability in handling complex features. Deeplabv3+ demonstrates relatively stable performance, yet its overall mIoU remains low and does not reach the effectiveness of the other methods. Regarding the F1-Score, the improved approach performs similarly to its mIoU results, consistently achieving the highest scores, especially in the later iterations, where the curve stabilizes and continues to rise. The F1-Score of Mask2Former is closely followed but experiences a slight decline in the middle iterations.

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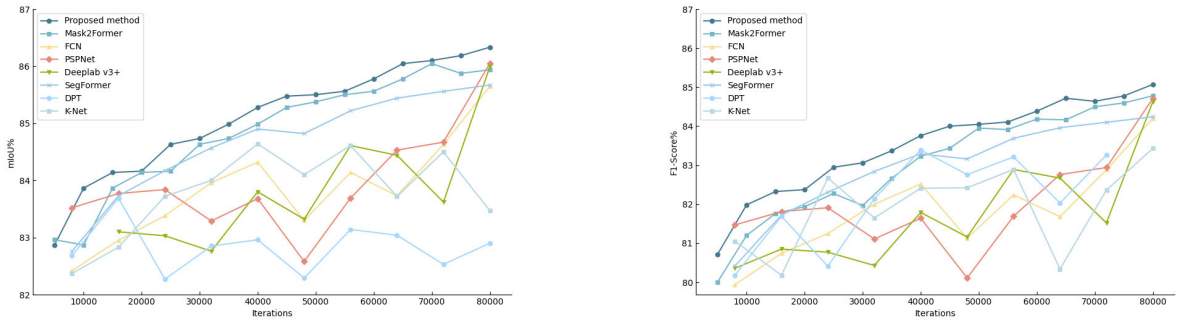

(a) mIoU growth of different methods (b) F1-Score growth of different methods

Fig. 6. Changing curves of evaluation metrics on the validation set during the comparative experiment training process.

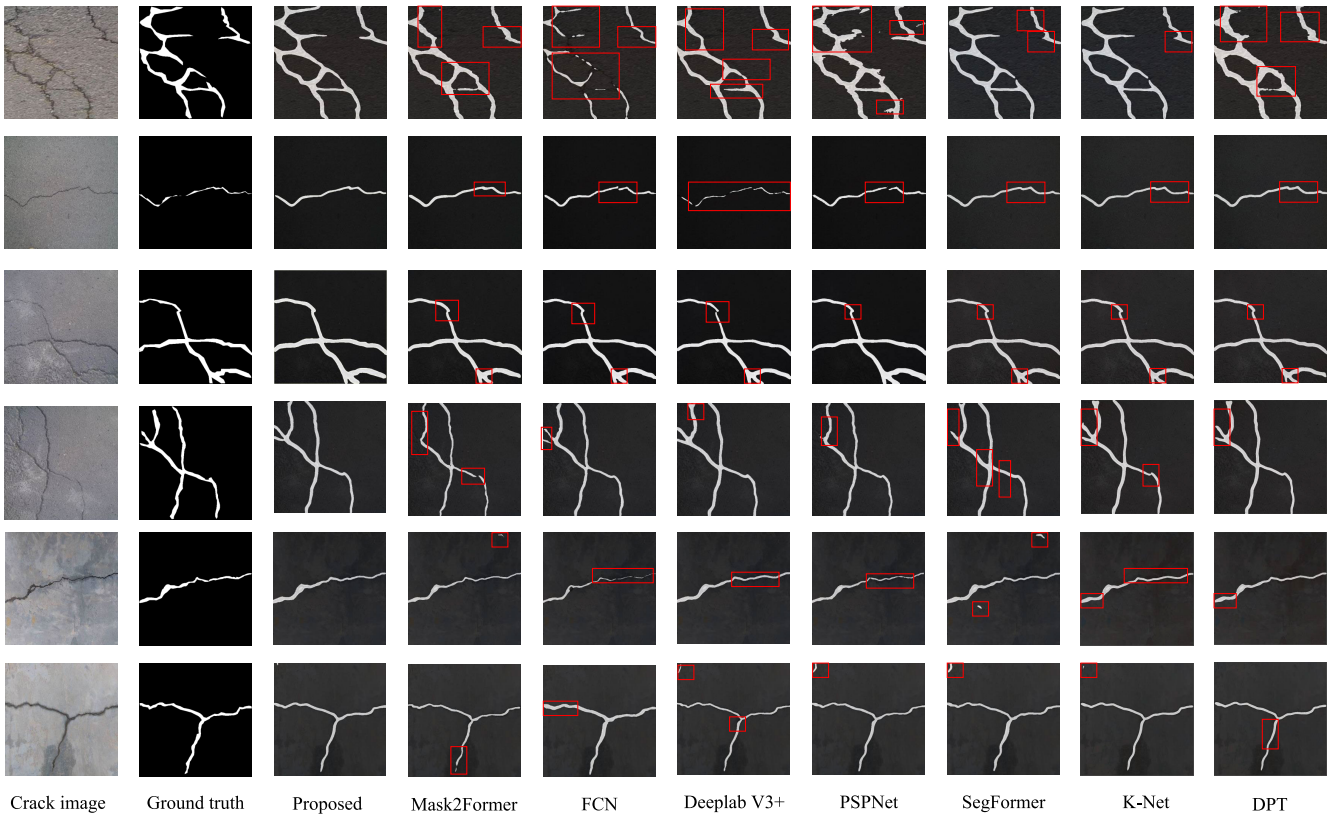

Fig. 7. Results of different methods in the test dataset.

The F1-Score curves of FCN and PSPNet show significant fluctuations, while the performance of Deeplabv3+ remains slightly lower. Although Deeplabv3+ improves in the later iterations, it still lags behind the other methods. Overall, the proposed method not only achieves continuous improvements in mIoU and F1-Score throughout the training process but also demonstrates superior stability. These results fully substantiate the advantages of the improved approach in semantic segmentation tasks. The model maintains high generalization ability, significantly enhancing both the accuracy and robustness of segmentation.

Fig. 7 presents the performance of various models in the crack segmentation task. Through comparison, it is evident that the improved algorithm excels particularly in crack detection, effectively capturing more details of cracks, especially along the edges and fine structures of smaller cracks. Its clarity and completeness surpass those of other methods. Compared to other models, the improved algorithm demonstrates superior performance in terms of both the accuracy and completeness of crack detection. While other models may miss some cracks in complex backgrounds, the improved algorithm is more precise in handling details and detecting edges, exhibiting higher

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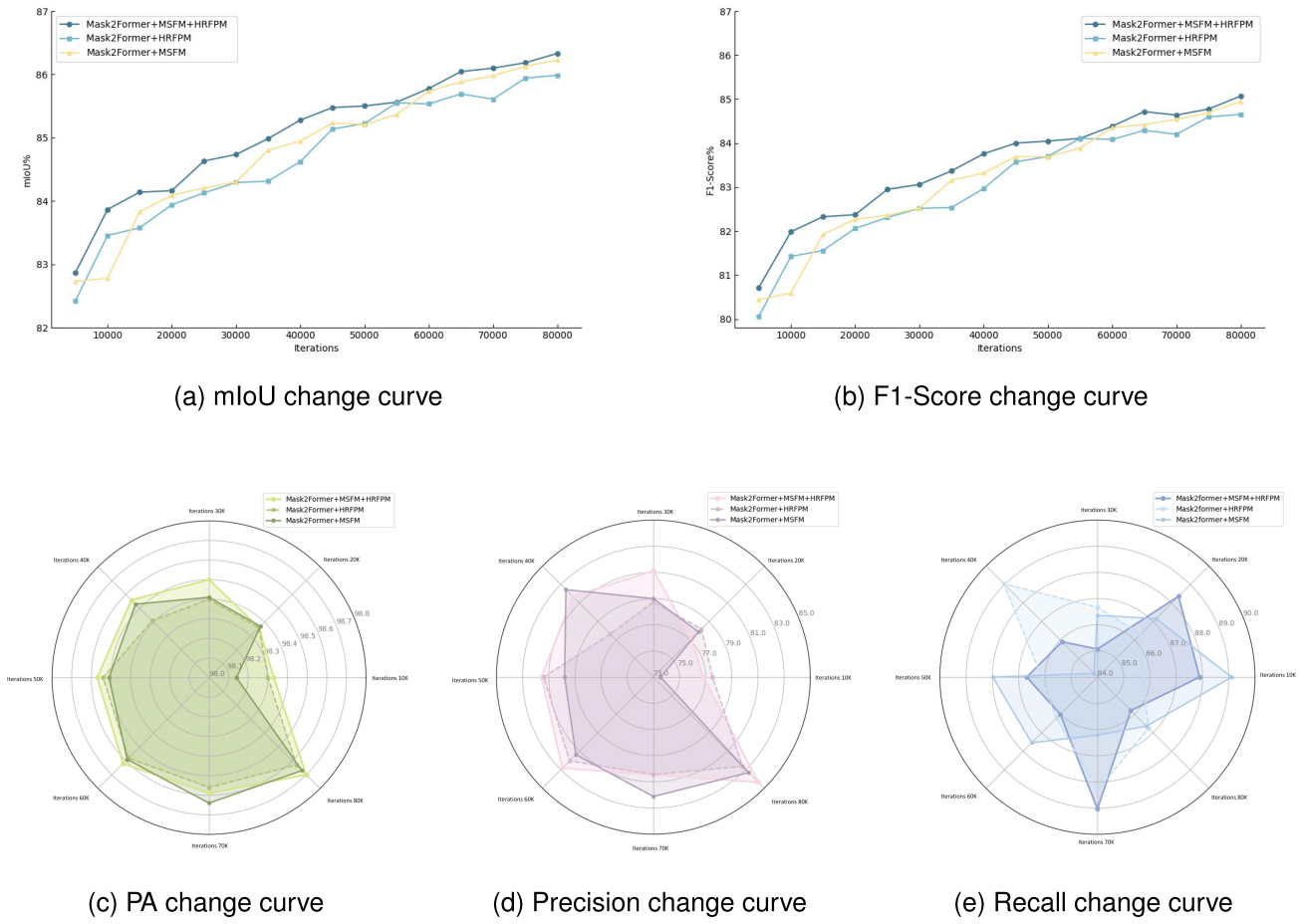
TABLE II
COMPARISON OF METRICS FOR DIFFERENT ALGORITHMS

Algorithms	Metrics					
	mIoU	PA	F1-Score	Precision	Recall	Parameters
Mask2Former	85.58	98.57	84.16	81.37	**87.14**	44.00M
Deeplabv3+	86.00	84.18	84.63	**85.08**	84.18	**26.68M**
PSPNet	86.05	85.21	84.71	84.22	85.21	46.71M
FCN	85.65	83.15	84.20	85.27	83.15	134.26M
SegFormer	85.67	85.52	84.24	83.00	85.52	54.80M
KNet	84.64	84.04	82.41	80.85	84.04	60.00M
DPT	83.14	84.15	83.21	83.64	83.18	492.00M
Proposed method	**86.33**	**98.70**	**85.07**	84.36	85.80	48.40M



(a) mIoU change curve (b) F1-Score change curve

(c) PA change curve (d) Precision change curve (e) Recall change curve

Fig. 8. Variation of model evaluation metrics with the number of iterations under different ablation conditions.

stability and robustness. The comparison with Ground Truth shows that the improved model restores the shape and position of cracks more accurately, further highlighting its advantage in crack detection tasks. In contrast, the detection results of this method are closer to the real annotations.

The experimental results presented in Table II reveal that the proposed improved algorithm demonstrates outstanding performance across several key metrics, particularly in terms of mIoU, PA, and F1-Score, surpassing the other compared algorithms. Specifically, the mIoU of the improved algorithm reaches 86.33%, the highest among all algorithms, indicating that it outperforms Mask2Former, Deeplabv3+, PSPNet, and FCN in overall segmentation performance. In terms of PA, the improved algorithm also leads with a maximum value of 98.70%, highlighting its significant advantage in accurately classifying pixels. Regarding the F1-Score, the improved algorithm again demonstrates superiority, achieving a score of 85.07%, slightly higher than the other algorithms, reflecting

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TABLE III
EVALUATION METRICS STATISTICS FOR ABLATION EXPERIMENT

Algorithms	Metrics				
	mIoU	PA	F1-Score	Precision	Recall
Mask2Former+HRFPM	85.99	98.63	84.65	82.64	**86.76**
Mask2Former+MSFM	86.22	98.67	84.95	83.32	86.64
Mask2Former+MSFM+HRFPM	**86.33**	**98.70**	**85.07**	**84.36**	85.80

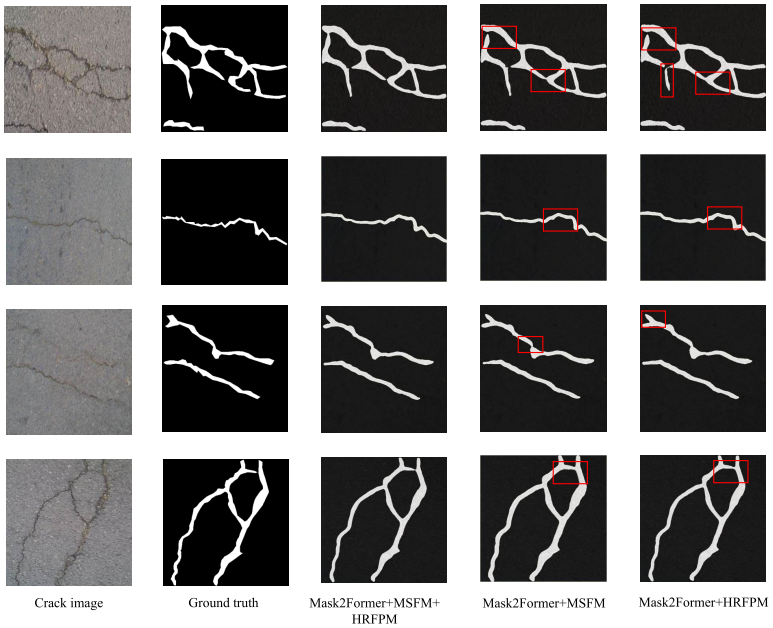

Fig. 9. Visualization of ablation experiment results.

better balance between precision and recall. Additionally, the recall rate of the improved algorithm reaches 85.80%, surpassing most of the compared algorithms, further validating its robustness in identifying positive examples. Although the precision is slightly lower than that of Deeplabv3+ and FCN, the overall performance remains at a relatively high level. The proposed method has 48.4M parameters, slightly more than Mask2Former (44M) and Deeplabv3+ (26.68M), but far fewer than FCN (134.26M), achieving a good balance between performance and model complexity. This suggests that, while maintaining lower model complexity, the improved algorithm is still capable of delivering remarkable performance across multiple crucial metrics, showcasing its significant advantage in the semantic segmentation of bridge cracks.

D. Analysis of the Ablation Experiment Results

This study verifies the effectiveness of the multi-scale feature fusion module and the high-resolution feature processing module through ablation experiments. As shown in the experimental results of Fig. 8, the Mask2Former model, which integrates multi-scale fusion and high-resolution features, achieves optimal performance in both mIoU and F1-Score. Throughout different training iterations, the mIoU and F1-Score values of this model steadily increase with continued training, consistently maintaining the highest levels. The model demonstrates a clear advantage in Precision and PA, with high stability, while the differences in Recall among the models are minimal. In contrast, models that use either high-resolution features or multi-scale fusion alone are less effective than the combined model, particularly in handling complex crack shapes and fine cracks. The experimental results indicate that the combination of multi-scale fusion and high-resolution features can more effectively capture various crack characteristics, enhancing the accuracy and robustness of segmentation results.

Fig. 9 presents the visualization results of the ablation experiment, comparing the performance of different modules in crack segmentation. Through these segmentation masks, the differences in crack edge detection and detail retention across various methods are visually apparent. The results show that the final approach, which combines the two modules, demonstrates clear superiority in overall performance. It not only captures fine cracks more accurately but also effectively reduces the occurrences of false detections and missed detections. Table III presents the statistical results of the evaluation metrics from the ablation experiment, which is used to compare the performance of different algorithms

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in the crack segmentation task. The final combined method achieves the best overall performance, with mIoU of 86.33%, PA of 98.70%, F1-Score of 85.07%, precision of 84.36%, and recall of 85.80%. The method integrating the two modules maintains a superior balance across all metrics. It enhances overall segmentation quality by capturing minute crack features more accurately, ensuring stable detection performance even in complex scenarios. This improvement makes crack detection more reliable.

V. Conclusion

This study proposes an improved bridge crack semantic segmentation algorithm based on an enhanced Mask2Former architecture. By integrating an MSFM and an HRFPM, the proposed method significantly improves segmentation continuity, robustness, and accuracy in detecting localized crack features. (1) MSFM dynamically integrates deep and shallow features using RepBlock, effectively balancing global semantic information and local details. HRFPM employs WTConv and DEConv to strengthen feature representation and incorporates CBAM to suppress background noise. The synergy of these modules enhances segmentation accuracy, particularly for fine crack edges. (2) Experimental results show the proposed method achieves an mIoU of 86.33%, PA of 98.70%, and an F1-Score of 85.07%, outperforming mainstream algorithms such as DeepLabv3+ and Segformer. Additionally, the model parameter count (48.4M) is significantly lower compared to complex networks like FCN (134.26M). Ablation studies further confirm the combined effectiveness, with the integrated model (86.33% mIoU) surpassing individual modules separately. (3) The proposed method accurately detects various types of cracks in complex backgrounds, effectively reducing false positives and missed detections from manual inspections, thus offering an efficient automated solution for structural health monitoring of bridges, demonstrating significant practical value. (4) Future research will focus on optimizing computational efficiency for real-time monitoring, expanding cross-scene datasets for enhanced generalization, and exploring Self-supervised learning and lightweight designs to further improve performance.

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